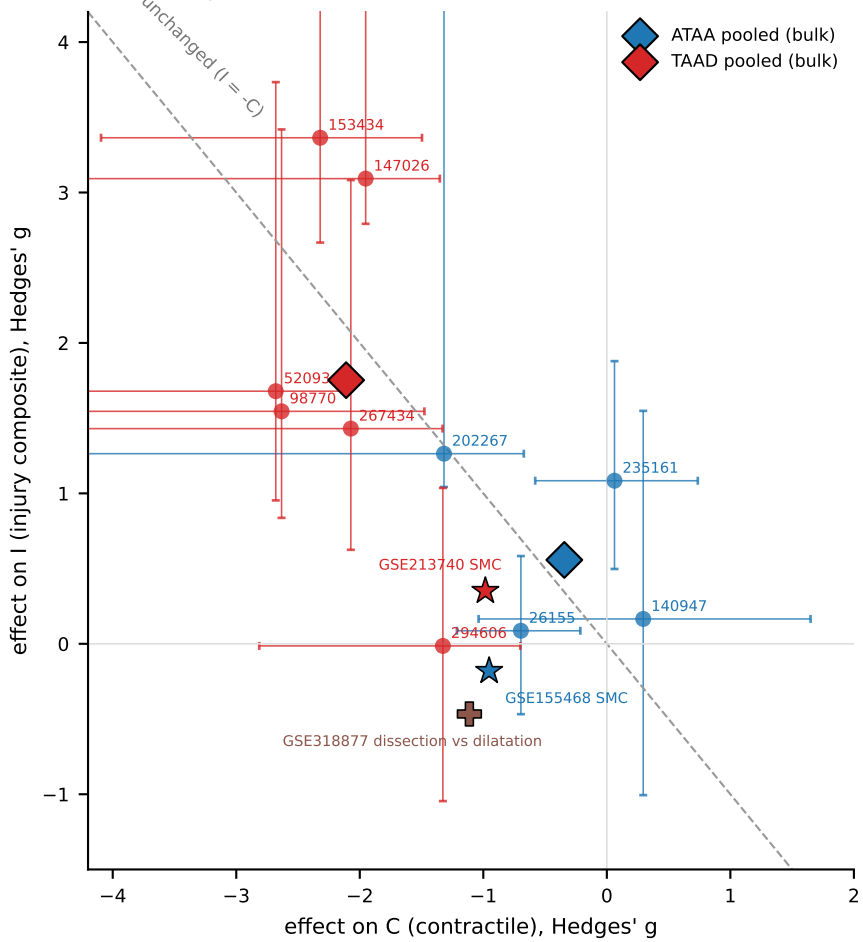


a Effects on the two programmes across datasets (circles: bulk cohorts; diamonds: REM pooled; stars: patient-level contractile SMC; cross: direct 3-vs-4 contrast)



b

programme (Tier 1)	ATAA vs control	TAAD vs control	differs between diseases?	within-cohort / patient-level evidence
C SMC contractile	-0.34	-2.11*	yes (Holm $P < 0.001$)	decreases with dilatation, GSE26155 (BH 0.02); lower in SMC of both scRNA sets (CI excl. 0; n small)
I injury composite	+0.56	+1.75*	borderline (Holm $P = 0.10$)	no trend with dilatation (GSE26155); SMC level: CI includes 0 in both scRNA sets
R relative state (I + C)	+0.24	+0.20	no ($P = 0.92$)	F2 and F3 primary tests not passed; GSE318877 estimation only
Glycolysis / oxidative / MYC-ribosome	+0.5 / +0.2	+1.8 / +1.5	yes (BH 0.01-0.02)	no trend with dilatation; attenuated after conditioning in ATAA only
IFN-alpha response	+0.80*	+0.08	no (BH 0.30)	increases with dilatation (BH 0.01)
MHC-II antigen presentation	+0.91*	-0.57	borderline (BH 0.09)	increases with dilatation (BH 0.01); higher in ATAA SMC (exact P 0.07; n small)
T-cell signature	+0.84*	+0.07	borderline (BH 0.08)	increases with dilatation (BH 0.01)
ECM-collagen	+0.60*	+0.05	no (BH 0.31)	increases with dilatation (BH 0.03)
Endothelial / fibroblast signatures	+0.3 / +0.3	-2.0 / -1.1	yes (BH 0.002 / 0.007)	composition level; not a within-cell finding

* pooled 95 % CI excludes 0. Tissue-level state comparison; no temporal or causal ordering is implied. Supported: cohort-level differences in C and in the metabolic/stress modules. Undetermined: whether the relative state R differs within the same wall cells (patient-level tests underpowered).